Beyond the marginal distribution: long-range correlations in prime gaps imprint on the 3D conformation of a self-avoiding polymer

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Abstract

The folding of a heteropolymer is governed by its monomer sequence, yet the intermediate regime between periodic and random sequences—those that are aperiodic but deterministic—remains largely unexplored. We introduce the Prime-Driven Topological Polymer (PDTP), a coarse-grained self-avoiding chain in which the local bending stiffness is set by an exponential map of the local prime gap, so that dense prime clusters act as flexible hinges and the long inter-prime runs as rigid rods. Using overdamped Langevin dynamics with simulated annealing, we ask whether the folded state is fixed by the composition of this sequence or by its arithmetic ordering. To separate the two, we compare the true prime chain against surrogates that preserve the marginal stiffness distribution exactly while destroying sequential order. The gross conformation is surrogate-invariant: neither the radius of gyration nor the tail-weighted core–shell amplitude distinguishes the true chain from a shuffle. The finer organization does: the Spearman correlation between stiffness and radial position is significantly suppressed for the true sequence ($\rho = +0.105$ vs. $+0.137$ at $N = 4 \times 10^4$; $p = 6.5 \times 10^{-5}$, $d = -0.96$), an effect a decomposition attributes to ordering rather than to the gap distribution. The imprint is quantitatively modest ($\sim 1\%$ of the radial-rank variance) but statistically robust and reproducible across system sizes, demonstrating that deterministic arithmetic order leaves a measurable trace on three-dimensional structure—one that survives the entropically favored radial sorting of the collapse.

1 Introduction

The folding of a heteropolymer into a compact, organized state is a canonical problem of soft-matter physics, underlying phenomena from protein structure to chromatin architecture [1, 2]. A primary determinant of the emergent morphology is the monomer sequence—the one-dimensional arrangement of chemically or mechanically distinct units along the backbone [3]. Two limiting classes of sequence have been studied in depth. At one extreme lie deterministic *periodic* sequences, epitomized by block copolymers, whose translational symmetry drives microphase separation into lamellar, cylindrical, and related ordered mesophases [4, 5]. At the other extreme lie *random* heteropolymers, in which the sequence is a quenched realization of uncorrelated disorder and the folding landscape is governed by frustration and glassy statistics [6, 7]. Between these poles lies a

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comparatively unexplored regime: sequences that are neither periodic nor random, but *aperiodic and deterministic*, carrying long-range correlations of purely structural origin. How such long-range-ordered aperiodicity is expressed in the three-dimensional conformation of a collapsed chain remains largely open.

Number theory has repeatedly furnished physics with precisely this kind of structured aperiodicity. The statistics of the nontrivial zeros of the Riemann zeta function coincide with the spectral fluctuations of random-matrix ensembles, forming a celebrated bridge between arithmetic and quantum spectra [8, 9]. In real space, aperiodic tilings and quasicrystals demonstrate that deterministic, non-periodic order can generate sharp diffraction and hierarchical organization absent from both crystals and glasses [10, 11]. The sequence of prime numbers occupies a distinctive place in this landscape: it is fully deterministic yet globally aperiodic, its coarse density decays only logarithmically as dictated by the prime number theorem, and its local structure—encoded in the sequence of prime gaps g_i, the distances between consecutive primes—carries well-documented correlations that are neither periodic nor memoryless [12, 13]. These properties make the prime-gap sequence a natural candidate for a designed, deterministic constraint to be mapped onto physical space, in the same spirit that quasiperiodic order has been imprinted onto photonic and mechanical metamaterials [14].

We realize this mapping mechanically. In the Prime-Driven Topological Polymer (PDTP), each bead i of a coarse-grained bead–spring chain is assigned a local bending stiffness that is an exponential function of its prime gap [Eq. (1)]. Dense clusters of primes (short gaps) become flexible hinges, while the rare long gaps become nearly rigid rods. Upon collapse, this construction establishes a competition between two tendencies. The first is generic to any stiff–flexible copolymer: excluded-volume repulsion and bending energy tend to expel rigid segments toward the periphery while flexible segments accommodate the curvature of the dense core, yielding a positive correlation between local stiffness and radial position [15, 16]. The second is specific to the arithmetic origin of the sequence: the deterministic ordering of the gaps along the backbone constrains which segments can reach the surface. The central question is whether this arithmetic ordering leaves any measurable imprint on the folded state, or whether the conformation is fixed entirely by the *marginal* distribution of stiffnesses, indifferent to their sequence. To resolve this we introduce a surrogate-based null model that preserves the marginal distribution exactly while destroying sequential order, and show—through a large ensemble of annealed trajectories—that the arithmetic ordering does leave a small but statistically robust signature, invisible in gross morphology and in marginal-matched nulls, yet detectable in the rank organization of the collapse.

2 Model definition

2.1 Prime skeleton and gap extraction

We identify the beads of a chain of length N with the integers $i = 0, 1, \ldots, N-1$. A sieve of Eratosthenes partitions these indices into primes and composites [12]. For a composite bead lying in the maximal run of consecutive composites bounded by the primes $p_{\text{left}} < i < p_{\text{right}}$ we define a *local prime gap* $g_i = p_{\text{right}} - p_{\text{left}}$, assigning every bead within a given inter-prime run the same value; for a prime bead $g_i = 0$. The boundary segments take $g_i = p_1$ for $i < p_1$ and $g_i = N - p_{\text{last}}$ for $i > p_{\text{last}}$. Within our largest system, $N = 4 \times 10^4$, the extremal gap is $g_{\max} = 72$, between the primes 31397 and 31469.

2.2 Gap-driven exponential stiffness

The gap field is mapped onto the local bending stiffness k_i through a saturated exponential map,

$$k_i = \begin{cases} k_{\text{hinge}}, & i \text{ prime}, \\ \min[\, k_{\text{hinge}}\, e^{\alpha g_i},\ k_{\max}\,], & i \text{ composite}. \end{cases} \tag{1}$$

The exponential rather than linear map is essential: because the gap envelope varies only slowly, a linear coupling produces a stiffness field too smooth to break the conformational symmetry, whereas the exponential map converts modest gap fluctuations into a stiffness contrast spanning nearly four orders of magnitude ($k_{\max}/k_{\text{hinge}} = 2000$ at our parameters). The amplification coefficient α and cap $k_{\max}$ are fixed by an independent stability calibration (Sec. 3).

2.3 Coarse-grained potential energy

Bonds between consecutive beads are harmonic,

$$E_{\text{bond}} = \tfrac{1}{2} k_{\text{bond}} \sum_{i=1}^{N-1} \big(|\mathbf{r}_i - \mathbf{r}_{i-1}| - b_0\big)^2, \tag{2}$$

with rest length b_0. The gap-driven stiffness enters through a cosine bending potential on interior triplets,

$$E_{\text{bend}} = \sum_{i=1}^{N-2} k_i\,(1-\cos\theta_i)\,, \qquad \cos\theta_i = \frac{(\mathbf{r}_i - \mathbf{r}_{i-1})\cdot(\mathbf{r}_{i+1} - \mathbf{r}_i)}{|\mathbf{r}_i - \mathbf{r}_{i-1}|\,|\mathbf{r}_{i+1} - \mathbf{r}_i|}, \tag{3}$$

so that k_i acts as a local, sequence-dependent bending modulus [15, 16]. Self-avoidance is a purely repulsive Weeks–Chandler–Andersen (WCA) interaction between non-bonded pairs ($|i-j| > 1$) [17],

$$E_{\text{WCA}}(r) = \begin{cases} 4\varepsilon\big[(\sigma/r)^{12} - (\sigma/r)^6\big] + \varepsilon, & r < r_c = 2^{1/6}\sigma, \\ 0, & r \geq r_c, \end{cases} \tag{4}$$

evaluated with a k-d-tree neighbor list and with the pairwise force magnitude capped at $f_{\text{cap}}^{\text{WCA}}$ as a numerical safeguard. Finally, a weak radial restoring force tethers each bead to the instantaneous center of mass $\mathbf{r}_{\text{cm}}$,

$$\mathbf{F}_i^{\text{conf}} = -\,k_{\text{conf}}\,(\mathbf{r}_i - \mathbf{r}_{\text{cm}}), \tag{5}$$

with k_{conf} small: it gathers the globule without crushing the rigid rods, allowing the gap-driven gradient to survive collapse.

2.4 Overdamped Langevin dynamics and annealing

The chain is propagated with overdamped (Brownian) Langevin dynamics in reduced units ($\sigma = \varepsilon = k_B = 1$) [18]:

$$\mathbf{r}_i \leftarrow \mathbf{r}_i + \mu\,\mathbf{F}_i\,\Delta t + \sqrt{2\mu\,k_B T\,\Delta t}\;\boldsymbol{\xi}_i, \qquad \boldsymbol{\xi}_i \sim \mathcal{N}(\mathbf{0}, \mathbb{I}_3), \tag{6}$$

where $\mathbf{F}_i = -\nabla_i E$, μ is the mobility, and $\boldsymbol{\xi}_i$ is independent Gaussian noise. Compaction is achieved by simulated annealing [19], lowering the temperature geometrically,

$$k_B T(t) = k_B T_{\text{hi}}\,(T_{\text{lo}}/T_{\text{hi}})^{t/(n_{\text{steps}}-1)}\,. \tag{7}$$

Per-bead displacements are capped at $|\Delta\mathbf{r}_i| \leq \delta_{\max}$, and each trajectory is initialized as a correlated random walk (persistence 0.55). Parameters are listed in Table 1.

Table 1: Model and protocol parameters (reduced units).

Symbol	Value	Meaning
b_0	1.0	bond rest length
k_{bond}	45.0	bond stiffness
k_{hinge}	0.25	baseline (prime) bending stiffness
α	0.15	gap amplification coefficient (calibrated)
$k_{\max}$	500.0	bending-stiffness cap (calibrated)
$\sigma,\ \varepsilon$	1.0, 1.0	WCA diameter, energy scale
r_c	$2^{1/6}\sigma$	WCA cutoff
$f_{\text{cap}}^{\text{WCA}}$	30.0	WCA force cap
k_{conf}	0.02	weak confinement constant
μ	1.0	mobility
Δt	0.008	integration step
$\delta_{\max}$	0.25	per-bead displacement cap
$k_BT_{\text{hi}},\ k_BT_{\text{lo}}$	0.50, 0.003	annealing temperature range

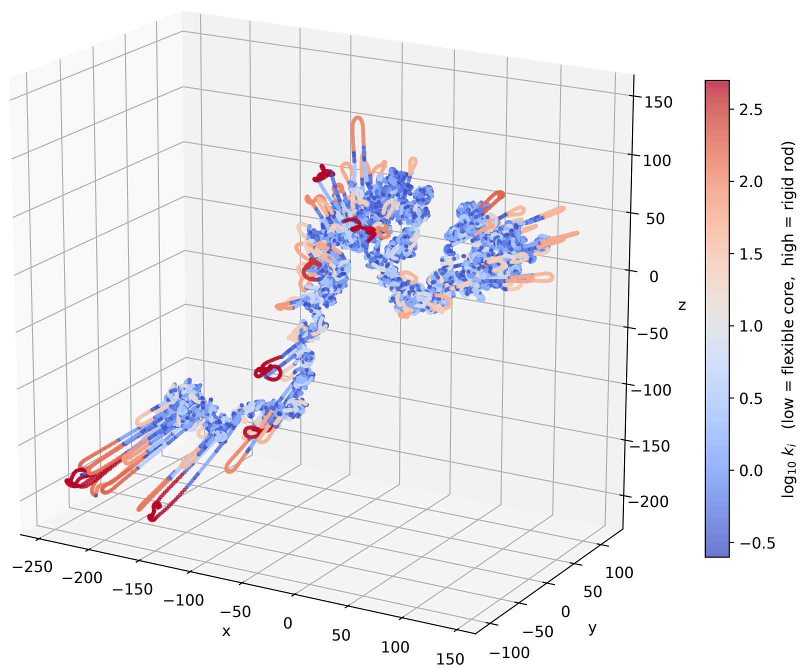

Figure 1: Representative folded conformation of the PDTP chain at $N = 4 \times 10^4$ (overdamped Langevin with simulated annealing, 3800 steps, seed 42). Beads are colored by $\log_{10}$ of the local bending stiffness k_i [Eq. (1)], from flexible prime-rich hinges (blue, $k_i = k_{\text{hinge}} = 0.25$) to rigid long-gap rods (red, capped at $k_{\max} = 500$). The chain collapses to a compact globule ($R_g = 110.7$) in which dense flexible cores are threaded by rigid rods extending into peripheral loops; for this conformation the stiffness–radius rank correlation is $\rho = +0.065$ and the tail-weighted core–shell separation $\Delta R = +21.0$.

3 Pre-flight calibration

Because the stiffness field of Eq. (1) is amplified exponentially, the rare long-gap sites carry bending moduli orders of magnitude above the baseline, and an explicit integrator can in principle diverge within a single step. We bound the parameters *a priori* through a worst-case stress test on an isolated segment [20]. We locate the global maximum gap ($g_{\max} = 72$, between the primes 31397 and 31469) and extract the 150-bead window centred on it (indices $[31358, 31508)$). The segment is evolved under bond and gap-driven bending forces alone, with the production displacement cap *removed* so that any latent divergence is exposed, while a small thermal noise ($k_B T = 0.05$) continually injects curvature.

Two findings emerge. First, the cosine bending potential of Eq. (3) is *self-limiting*: its restoring force scales as $\sin\theta_i$ and vanishes as a segment straightens, so a rigid rod relaxes toward the straight configuration without overshoot. Sweeping $\alpha \in [0.05, 0.15]$—and, independently, relaxing the cap to the fully uncapped limit, where the stiffness at $g_{\max}$ reaches $k \approx 1.2 \times 10^4$ (nearly $5 \times 10^4\, k_{\text{hinge}}$)—produces no divergence over the test window. The exponential amplification is thus *not* the binding stability constraint. Second, the binding constraint is set by the stiffest *harmonic* term, the bonds: sweeping the timestep, the scheme is stable for $\Delta t \le 0.01$ but diverges within a few steps for $\Delta t \gtrsim 0.02$, consistent with the fastest bond-mode limit $\Delta t \lesssim 2/(4\mu k_{\text{bond}}) \approx 0.011$. Guided by this, the production runs adopt $\alpha = 0.15$ with a conservative cap $k_{\max} = 500$ (a factor ≈ 25 below the uncapped worst case) and a hard displacement bound $\delta_{\max} = 0.25$; with $\Delta t = 0.008$ lying a factor ≈ 2.5 below the divergence threshold, all trajectories operate well inside the stable regime. The calibration's purpose is not to maximize the stiffness contrast but to certify that the symmetry-breaking map can be integrated without numerical artifacts that could masquerade as structural signal.

4 Results

We compare the true prime chain against surrogates under identical force fields, dynamics, and annealing; the only difference between arms is the anchor pattern generating the stiffness field. Three observables characterize the collapsed state: the radius of gyration R_g; the percentile core–shell separation ΔR, the mean radial distance of the stiffest 20% of beads minus that of the most flexible 20%; and the Spearman rank correlation ρ between local stiffness k_i and radial distance. Positive ρ or ΔR indicates rigid segments at the periphery. Ensembles comprise n independent trajectories per arm.

4.1 Global collapse is surrogate-invariant

All arms collapse along indistinguishable trajectories and their gross dimensions do not separate. At $N = 10^4$ the radius of gyration is $R_g = 57 \pm 16$ (true) versus 49 ± 13 (shuffle) (Welch $p = 0.20$, $n = 10$), and the gross core–shell amplitude is likewise indistinguishable: $\Delta R = 14.4 \pm 0.7$ versus 15.1 ± 0.5 at $N = 4 \times 10^4$ ($p = 0.37$, $n = 40$). The overall size and the gross magnitude of stiffness segregation are therefore fixed by the marginal stiffness distribution, which every surrogate preserves.

4.2 A core–shell ordering signature

The rank correlation ρ reveals a signature of ordering. In our decisive experiment—$N = 4\times10^4$, deep annealing (8000 steps), $n = 40$ per arm—both arms exhibit positive ρ, but the sorting is significantly *weaker* for the true chain: $\rho = +0.105 \pm 0.007$ versus $+0.137 \pm 0.004$ (Welch $p = 6.5 \times 10^{-5}$, Mann–Whitney $p = 3.1 \times 10^{-4}$, $d = -0.96$; Fig. 2). The deterministic ordering thus imposes a reproducible *resistance* to the radial sorting generic stiff–flexible physics would otherwise produce. The effect is quantitatively modest—stiffness accounts for only $\rho^2 \approx 1\%$ of the radial-rank variance—but statistically robust. The same comparison in ΔR is not resolved at this size ($p = 0.37$): the ordering signal resides in the global rank structure, not in the stiffness extremes, which is why the low-variance estimator ρ is required.

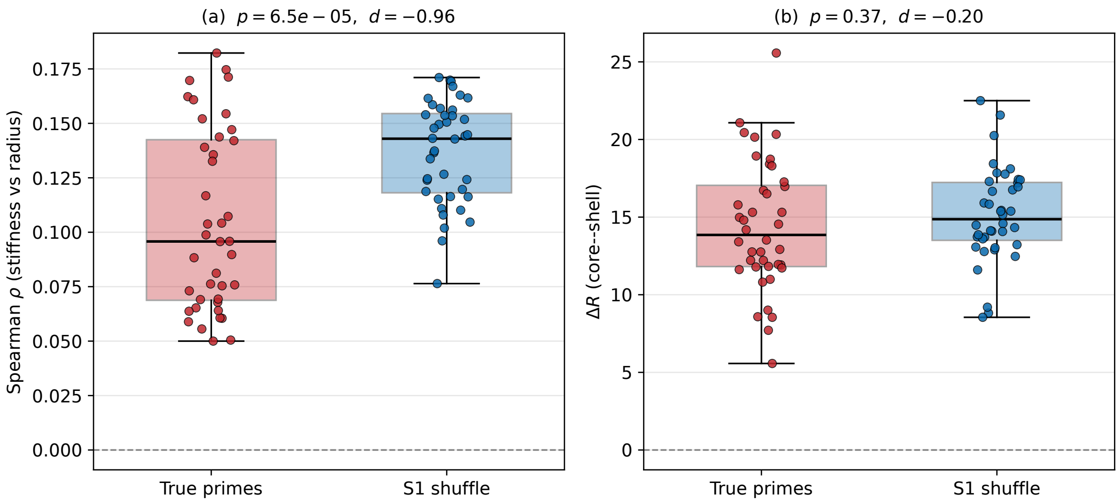

Figure 2: Decisive test at $N = 4 \times 10^4$, deep annealing (8000 steps), $n = 40$ per arm. (a) Spearman ρ: true primes $+0.105 \pm 0.007$ vs. shuffle S1 $+0.137 \pm 0.004$ (Welch $p = 6.5 \times 10^{-5}$, Mann–Whitney $p = 3.1 \times 10^{-4}$, $d = -0.96$)—the true sequence is significantly less radially sorted. (b) $\Delta R = 14.4 \pm 0.7$ vs. 15.1 ± 0.5 ($p = 0.37$, $d = -0.20$), statistically indistinguishable. Boxes: median and quartiles; points: individual runs.

4.3 Ordering versus marginal distribution

To attribute the effect we compare three arms at $N = 10^4$, $n = 40$: the true chain; the Shuffle (S1), preserving the exact gap multiset but randomizing order; and a Cramér surrogate (S3), placing pseudo-primes at density $1/\ln i$, preserving only the coarse density trend [13]. The decomposition is clean (Fig. 3). The ordering contrast (true vs. S1) is strongly significant ($\Delta\rho = -0.051$, $p = 1.6 \times 10^{-5}$, $d = -1.05$). The marginal-shape contrast (S1 vs. S3), both order-randomized, is small and not significant ($\Delta\rho = -0.017$, $p = 0.09$, $d = -0.38$). The combined contrast (true vs. S3) is largest ($\Delta\rho = -0.068$, $p = 1.2 \times 10^{-6}$, $d = -1.18$). Arithmetic ordering therefore accounts for the bulk of the effect ($\approx 75\%$ of the combined shift), while the shape of the gap distribution contributes little. This is the central result: what the folded conformation registers is not the composition of the prime-gap sequence but its sequential, long-range arithmetic structure.

4.4 Finite-size behaviour

Finally, we examine the size dependence across $N = 10^4, 2 \times 10^4, 4 \times 10^4$ (Fig. 4). In both ρ and ΔR, the true chain lies systematically below the shuffle at every size, with significance at $N = 10^4$ and 2×10^4 (ρ: $p = 3 \times 10^{-3}$ and 2×10^{-3}). An initial sweep at $N = 4 \times 10^4$ with reduced

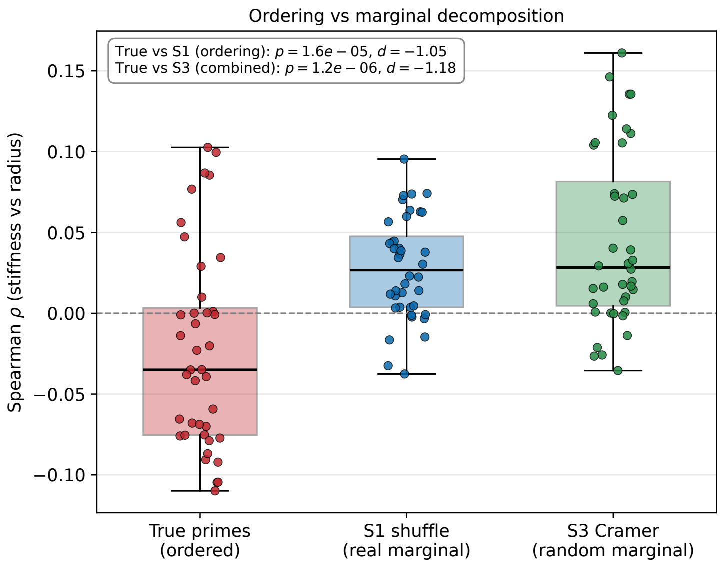

Figure 3: Ordering versus marginal-distribution decomposition at $N = 10^4$, $n = 40$ per arm. True primes ($\rho = -0.023 \pm 0.010$), shuffle S1 ($+0.028 \pm 0.005$; exact gap multiset, order randomized), and Cramér surrogate S3 ($+0.045 \pm 0.009$; pseudo-primes at density $1/\ln i$). Ordering (true vs. S1): $p = 1.6 \times 10^{-5}$, $d = -1.05$. Marginal shape (S1 vs. S3): $p = 0.09$ (n.s.). Combined (true vs. S3): $p = 1.2 \times 10^{-6}$, $d = -1.18$.

statistics ($n = 20$) and shallower annealing returned a gap below threshold ($p = 0.075$); the decisive experiment of Sec. 4.2 shows this to be a power-and-protocol artifact rather than a genuine fade—at matched per-bead annealing depth and $n = 40$, the same gap is recovered at $p = 6.5 \times 10^{-5}$. We caution that the *absolute* correlation grows with N for both arms (from $\rho \approx 0$ at $N = 10^4$ to $\rho \approx +0.1$ at $N = 4 \times 10^4$), reflecting both progressive collapse and the varying annealing depth of the sweep; consequently only the fixed-N gap between true and shuffle constitutes a controlled comparison, and we read the absolute N-dependence as a trend rather than an equilibrium scaling law. Within the range studied, once statistics and annealing depth are controlled, the ordering signature is present and significant at every size.

5 Discussion

5.1 What the fold registers

The representative conformation (Fig. 1; $R_g \approx 111$) displays the generic architecture of a semiflexible heteropolymer: dense flexible cores threaded and flanked by rigid rods that extend into loops toward the periphery. Both the true chain and its surrogates share this gross morphology; their difference lies elsewhere. Neither R_g nor ΔR distinguishes the true sequence from a stiffness-matched shuffle; only the rank correlation ρ does, and there the true sequence is significantly *less* sorted. The folded conformation thus behaves as a physical readout sensitive not to the composition of the stiffness sequence but to its sequential arrangement—specifically to correlations that a random permutation destroys.

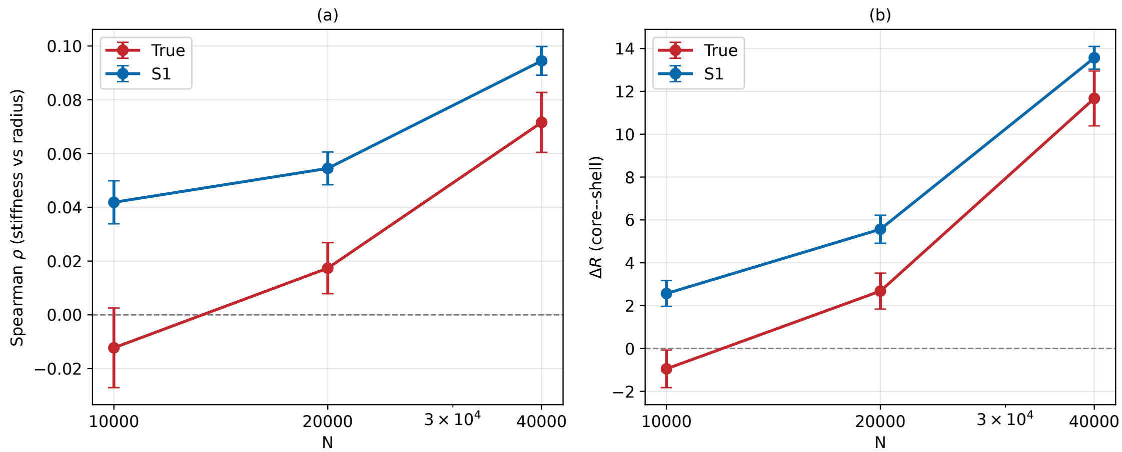

Figure 4: Finite-size behaviour, true vs. shuffle S1, for $N = 10^4, 2 \times 10^4, 4 \times 10^4$ ($n = 20$ per arm; annealing 2000/3500/5000 steps). (a) Spearman ρ and (b) ΔR vs. N (mean $\pm$ s.e.m.). The true chain lies below the shuffle at every size. The gap is significant at $N = 10^4$ and 2×10^4; the reduced-statistics point at $N = 4 \times 10^4$ ($p = 0.075$) is resolved by the deeper-annealed, higher-n experiment of Fig. 2 ($p = 6.5 \times 10^{-5}$). Only the fixed-N true–shuffle gap is a controlled comparison.

5.2 A candidate mechanism

Why should ordering matter once the chain has collapsed? We advance a concrete, falsifiable hypothesis. In the true sequence the long gaps are not placed independently: prime gaps carry documented correlations and clustering [21, 22], superposed on a weak positional drift. We conjecture that this correlated placement embeds a fraction of the rigid rods within flexible neighborhoods, impeding their migration to the surface and thereby *reducing* the stiffness–radius correlation relative to a sequence in which the identical rods are scattered independently. Concretely, the abundance of small gaps —twin primes (gap 2) alone account for ≈14% of all gaps at $N = 4 \times 10^4$—creates local clusters of flexible hinges [21], which we hypothesize act as *topological traps* that anchor neighboring rigid rods within the interior and oppose their radial expulsion. The hypothesis is directly testable with scale-selective surrogates—for instance a block-preserving permutation that retains local gap order within contiguous segments while scrambling their large-scale arrangement, and its converse—which would localize the correlation length responsible for the signal. The present data establish that ordering matters, not yet how.

5.3 Connections

Our result offers a physical manifestation of the fact that the prime-gap sequence is neither periodic nor memoryless [12, 21, 22]: a folding process, despite the drastic coarse-graining it imposes, still transmits a residue of arithmetic order into three-dimensional structure. In this sense the polymer acts as an unconventional probe of long-range correlations in the primes, complementary to spectral and combinatorial approaches [8, 9]. On the physical side, the construction sits within the study of aperiodic and deterministic order—quasicrystals, aperiodic tilings, and sequence-programmed matter [10, 11, 14]. We are, however, cautious about extrapolation: the effect is small in absolute terms, and we make no claim of a route to functional metamaterials. What we demonstrate is narrower and cleaner: that a deterministic, aperiodic arithmetic sequence leaves a detectable, reproducible imprint on the equilibrium-like conformation of a physical model, distinguishable from every marginal-matched null.

5.4 Limitations

Several limitations bound these conclusions. First, the effect is quantitatively small: local stiffness accounts for only $\sim 1\%$ of the radial-rank variance, the distributions overlap heavily, and no single conformation can be classified as "prime" or "shuffled"—the signal is a shift in the ensemble mean. Second, our ensembles are generated by simulated annealing rather than sampled from a true equilibrium distribution; absolute observables depend on the protocol, and only the fixed-N true–surrogate contrast, annealed identically, is controlled. Third, we have examined a single stiffness map—one α, one functional form, one confinement; robustness across this parameter space remains to be charted. Fourth, our sizes reach only $N = 4 \times 10^4$; the absolute correlation drifts upward with N, and while the gap is significant at every size once statistics and annealing depth are controlled, its fate in the thermodynamic limit is genuinely open. Finally, the signal is carried by the sensitive rank estimator ρ and is not resolved by the tail-based ΔR at the largest size. We emphasize that this is a controlled in-silico construction: we make no claim that prime numbers are physically privileged in nature, and none of biological relevance.

6 Conclusion

We introduced the PDTP model and used it to ask whether the folded state of an aperiodic-but-deterministic sequence is determined by its composition alone or also by its arithmetic ordering. By comparing the true prime chain against surrogates that preserve the marginal stiffness distribution exactly while destroying order, we found the answer to be affirmative but bounded. The gross conformation is surrogate-invariant—neither R_g nor ΔR distinguishes true from shuffle—so overall size and coarse morphology are fixed by the marginal distribution. The finer organization is not: the stiffness–radius rank correlation is significantly suppressed for the true sequence ($\rho = +0.105$ vs. $+0.137$ at $N = 4 \times 10^4$; $p = 6.5 \times 10^{-5}$, $d = -0.96$), an effect a three-way decomposition attributes to arithmetic ordering rather than to the gap distribution, and reproducible across sizes once statistics and annealing depth are controlled. The effect is quantitatively modest ($\sim 1\%$ of the radial-rank variance) and is a property of the ensemble rather than of any single fold. Beyond the specific case of primes, the PDTP framework and its surrogate-null methodology offer a controlled testbed for how one-dimensional deterministic order is—and is not—expressed in three-dimensional structure. The most immediate directions are to resolve the correlation scale responsible for the signal through scale-selective surrogates, to map the effect across the stiffness-map parameter space, and to determine its fate at system sizes well beyond those reached here.

Data and code availability

All code, data, and figure-generation scripts that reproduce the results of this paper are openly available at `https://github.com/Ruqing1963/prime-driven-topological-polymer`. The repository contains the pre-flight calibration, the surrogate/null-model decomposition, the finite-size sweep, the decisive $N = 4 \times 10^4$ experiment, and the conformation generator, together with the ensemble result files and a script that regenerates all figures.

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